

Contents

1	Introduction	1
2	Introduction2	2
3	Introduction 3	3
4	Supplementary Tables	4
4.1	Supplementary Table 1	4
4.2	Supplementary Table 2	4
4.3	Supplementary Table 3	4
4.4	Supplementary Table 4	4

1 Introduction

[1] Introduction Sex differences in the presentation and development of complex traits and diseases have been widely observed. These differences may arise from varying environmental exposures and/or a sex-specific genetic architecture characterized by differential gene regulation and other biological mechanisms. Understanding these sex differences is crucial for improving prevention, diagnosis, and treatment of a variety of human diseases. For instance, individuals with schizophrenia exhibit a fourfold higher prevalence of type 2 diabetes compared to the general population, with some risk potentially attributable to shared genetic etiology. Previous studies have shown sex-differences at SNP effect level (Liu J et al 2012 and 2013). Bernabeu et al. 2021 have addressed the sex-differential genetic architecture across a large number of well-powered sex-stratified traits from the UK Biobank. The study has found sex differences in both individual SNP effect and global (polygenic) SNP heritability, identifying 53 phenotypes with significant genetic architecture differences between males and females (Ishigaki, 2021). The complexity of studying genetic architecture is further compounded by pleiotropy, where genetic variants can influence multiple traits independently. A similar study conducted with BioBank Japan Project (BBJ) data, also found instances of SNP effect sex differences and further sex differences in the functional enrichment of SNP heritability (Ishigaki, 2020). Some studies have further identified sex differences in important cross-trait relationships within the domain of neurological diseases and related traits at the level of genetic correlations (Martin, 2021; Blockland, 2021). The existence of sex-differential selection and sexual antagonistic loci has been shown in UKBBdata (Ruzicka, 2022), pointing to sex differences in selection as an intriguing probable mechanism contributing to the sex differential genetic architecture of complex traits though pleiotropy. In line with this idea, some studies have shown enrichment in sex-differentiated SNP of anthropometric traits among top associated SNPs of some neurological traits, suggesting that the same mechanisms of secondary sex characteristics may also contribute to unrelated complex traits (Mitra (2016)). Some studies have begun to shed light onto the role of pleiotropic sex differences in the shaping of the relationships across seemingly unrelated traits and diseases. Putative shared etiology across seemingly unrelated traits and diseases as determined by genetic correlations has also been subjected to sex difference scrutiny with interesting results. For example, a male-specific significant genetic correlation exists between headache and T2D (Islam, 2022). Some studies have further addressed the nature of sex-differential genetic correlations and found sex-differences in causal relationships among traits, identifying sex differences in the causal association of several anthropometric traits to vulnerability to PTSD (Carvalho, 2021). This study is aimed to extend the study of pleiotropic sex differences across all available sex-stratified GWAS data and promises to yield a more comprehensive understanding of the pleiotropic genetic architecture of human traits and diseases. This project aims to extend previous work on the sex differences in the genetic architecture of human health-related traits and diseases by adopting a more comprehensive and robust approaches The proposed study will address previous limitations by focusing on three key extensions: By leveraging other large-scale publicly available datasets besides UK Biobank, this project will expand to the examination of more diseases, which are not generally well covered in UKBB. With a focus on pleiotropic sex-differences the project will address the sex-differences on the aetiological / causal relationships among traits. In addition to genome-

wide sex-differences, sex differences in locus-level are analysed to uncover subtle, region-specific genetic relationships that may go unnoticed in genome-wide level analyses. In conclusion, this project aspires to deepen our understanding of the intricate interplay between sex and genetics in human health and disease. By addressing the limitations of previous studies and employing new methodologies, the research will yield valuable insights into developing sex-specific diagnostic and therapeutic strategies for a wide range of health conditions.

2 Introduction2

[1] Introduction Sex differences in the presentation and development of complex traits and diseases have been widely observed. These differences may arise from varying environmental exposures and/or a sex-specific genetic architecture characterized by differential gene regulation and other biological mechanisms. Understanding these sex differences is crucial for improving prevention, diagnosis, and treatment of a variety of human diseases. For instance, individuals with schizophrenia exhibit a fourfold higher prevalence of type 2 diabetes compared to the general population, with some risk potentially attributable to shared genetic etiology. Previous studies have shown sex-differences at SNP effect level (Liu J et al 2012 and 2013). Bernabeu et al. 2021 have addressed the sex-differential genetic architecture across a large number of well-powered sex-stratified traits from the UK Biobank. The study has found sex differences in both individual SNP effect and global (polygenic) SNP heritability, identifying 53 phenotypes with significant genetic architecture differences between males and females (Ishigaki, 2021). The complexity of studying genetic architecture is further compounded by pleiotropy, where genetic variants can influence multiple traits independently. A similar study conducted with BioBank Japan Project (BBJ) data, also found instances of SNP effect sex differences and further sex differences in the functional enrichment of SNP heritability (Ishigaki, 2020). Some studies have further identified sex differences in important cross-trait relationships within the domain of neurological diseases and related traits at the level of genetic correlations (Martin, 2021; Blockland, 2021). The existence of sex-differential selection and sexual antagonistic loci has been shown in UKBBdata (Ruzicka, 2022), pointing to sex differences in selection as an intriguing probable mechanism contributing to the sex differential genetic architecture of complex traits through pleiotropy. In line with this idea, some studies have shown enrichment in sex-differentiated SNP of anthropometric traits among top associated SNPs of some neurological traits, suggesting that the same mechanisms of secondary sex characteristics may also contribute to unrelated complex traits (Mitra (2016)). Some studies have begun to shed light onto the role of pleiotropic sex differences in the shaping of the relationships across seemingly unrelated traits and diseases. Putative shared etiology across seemingly unrelated traits and diseases as determined by genetic correlations has also been subjected to sex difference scrutiny with interesting results. For example, a male-specific significant genetic correlation exists between headache and T2D (Islam, 2022). Some studies have further addressed the nature of sex-differential genetic correlations and found sex-differences in causal relationships among traits, identifying sex differences in the causal association of several anthropometric traits to vulnerability to PTSD (Carvalho, 2021). This study is aimed to extend the study of pleiotropic sex differences across all available sex-stratified GWAS data and promises to yield a more comprehensive understanding of the pleiotropic genetic architecture of human traits and diseases. This project aims to extend previous work on the sex differences in the genetic architecture of human health-related traits and diseases by adopting a more comprehensive and robust approaches. The proposed study will address previous limitations by focusing on three key extensions: By leveraging other large-scale publicly available datasets besides UK Biobank, this project will expand to the examination of more diseases, which are not generally well covered in UKBB. With a focus on pleiotropic sex-differences the project will address the sex-differences on the aetiological / causal relationships among traits. In addition to genome-wide sex-differences, sex differences in locus-level are analysed to uncover subtle, region-specific genetic relationships that may go unnoticed in genome-wide level analyses. In conclusion, this project aspires to deepen our understanding of the intricate interplay between sex and genetics in human health and disease. By addressing the limitations of previous studies and employing new methodologies, the research will yield valuable insights into developing sex-specific diagnostic and therapeutic strategies for a wide range of health conditions.

3 Introduction 3

[1] Introduction Sex differences in the presentation and development of complex traits and diseases have been widely observed. These differences may arise from varying environmental exposures and/or a sex-specific genetic architecture characterized by differential gene regulation and other biological mechanisms. Understanding these sex differences is crucial for improving prevention, diagnosis, and treatment of a variety of human diseases. For instance, individuals with schizophrenia exhibit a fourfold higher prevalence of type 2 diabetes compared to the general population, with some risk potentially attributable to shared genetic etiology. Previous studies have shown sex-differences at SNP effect level (Liu J et al 2012 and 2013). Bernabeu et al. 2021 have addressed the sex-differential genetic architecture across a large number of well-powered sex-stratified traits from the UK Biobank. The study has found sex differences in both individual SNP effect and global (polygenic) SNP heritability, identifying 53 phenotypes with significant genetic architecture differences between males and females (Ishigaki, 2021). The complexity of studying genetic architecture is further compounded by pleiotropy, where genetic variants can influence multiple traits independently. A similar study conducted with BioBank Japan Project (BBJ) data, also found instances of SNP effect sex differences and further sex differences in the functional enrichment of SNP heritability (Ishigaki, 2020). Some studies have further identified sex differences in important cross-trait relationships within the domain of neurological diseases and related traits at the level of genetic correlations (Martin, 2021; Blockland, 2021). The existence of sex-differential selection and sexual antagonistic loci has been shown in UKBBdata (Ruzicka, 2022), pointing to sex differences in selection as an intriguing probable mechanism contributing to the sex differential genetic architecture of complex traits through pleiotropy. In line with this idea, some studies have shown enrichment in sex-differentiated SNP of anthropometric traits among top associated SNPs of some neurological traits, suggesting that the same mechanisms of secondary sex characteristics may also contribute to unrelated complex traits (Mitra (2016)). Some studies have begun to shed light onto the role of pleiotropic sex differences in the shaping of the relationships across seemingly unrelated traits and diseases. Putative shared etiology across seemingly unrelated traits and diseases as determined by genetic correlations has also been subjected to sex difference scrutiny with interesting results. For example, a male-specific significant genetic correlation exists between headache and T2D (Islam, 2022). Some studies have further addressed the nature of sex-differential genetic correlations and found sex-differences in causal relationships among traits, identifying sex differences in the causal association of several anthropometric traits to vulnerability to PTSD (Carvalho, 2021). This study is aimed to extend the study of pleiotropic sex differences across all available sex-stratified GWAS data and promises to yield a more comprehensive understanding of the pleiotropic genetic architecture of human traits and diseases. This project aims to extend previous work on the sex differences in the genetic architecture of human health-related traits and diseases by adopting a more comprehensive and robust approaches. The proposed study will address previous limitations by focusing on three key extensions: By leveraging other large-scale publicly available datasets besides UK Biobank, this project will expand to the examination of more diseases, which are not generally well covered in UKBB. With a focus on pleiotropic sex-differences the project will address the sex-differences on the aetiological / causal relationships among traits. In addition to genome-wide sex-differences, sex differences in locus-level are analysed to uncover subtle, region-specific genetic relationships that may go unnoticed in genome-wide level analyses. In conclusion, this project aspires to deepen our understanding of the intricate interplay between sex and genetics in human health and disease. By addressing the limitations of previous studies and employing new methodologies, the research will yield valuable insights into developing sex-specific diagnostic and therapeutic strategies for a wide range of health conditions.

4 Supplementary Tables

4.1 Supplementary Table 1

4.2 Supplementary Table 2

4.3 Supplementary Table 3

4.4 Supplementary Table 4

References

- [1] Marian Beekman, Christa Nederstigt, H Eka D Suchiman, Dennis Kremer, Ruud van der Breggen, Nico Lakenberg, Wendimagegn Ghidye Alemayehu, Anton JM de Craen, Rudi GJ Westendorp, Dorret I Boomsma, et al. Genome-wide association study (gwas)-identified disease risk alleles do not compromise human longevity. *Proceedings of the National Academy of Sciences*, 107(42):18046–18049, 2010.